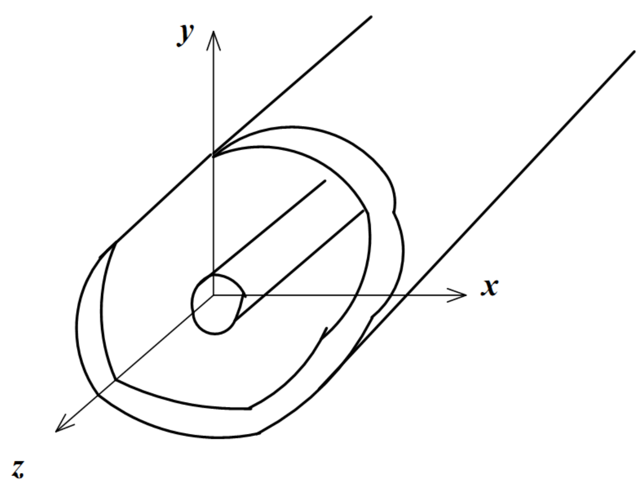
Waveguides

Guided propagation

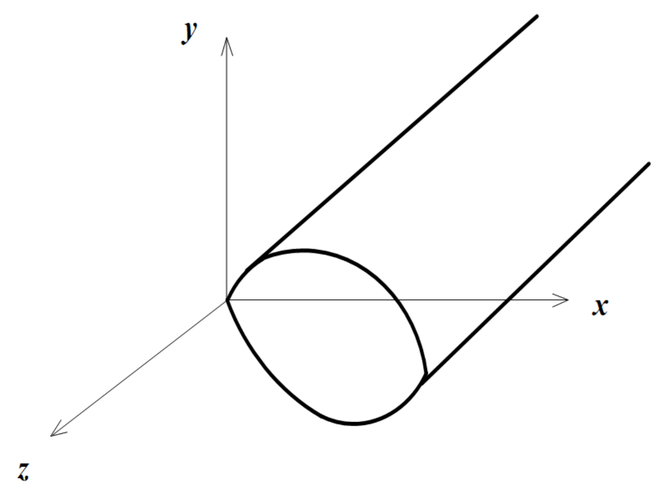
Wave propagation in coaxial cables is easy to understand, and can be derived from simple circuit laws (see Telegrapher's equations).

For very high frequencies, coaxial cables become impractical due to high losses and difficulty in constructing circuits with well-defined dimensions.

In this section we study the general theory of wave propagation, including the case where there is only one conductor in the problem.



a) General two conductor transmission line.



b) Closed Waveguide.

Maxwell's equations

Consider a source-free region of space, i.e. with no charges ($\rho = 0$) and no currents ($\boldsymbol{j} = 0$), filled by a non-conducting medium ($\sigma = 0$) having permittivity ϵ and permeability μ. Maxwell's equations are:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \qquad (35a)$$

$$\nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} \qquad (35b)$$

$$\nabla \cdot \mathbf{E} = 0 \qquad (35c)$$

$$\nabla \cdot \mathbf{H} = 0 \qquad (35d)$$

Combine (35a) and (35b) by taking

$$\nabla \times \nabla \times \mathbf{E} = -\mu \frac{\partial}{\partial t}(\nabla \times \mathbf{H}) = -\mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Since $\nabla \times \nabla \times \mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}$, the above becomes:

$$\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Wave equations

Calling $v = \frac{1}{\sqrt{\mu\epsilon}}$ the speed of light in the medium, we obtain:

$$\boxed{\nabla^2 \mathbf{E} = \frac{1}{v^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \qquad (36a)}$$

Analogously:

$$\boxed{\nabla^2 \mathbf{H} = \frac{1}{v^2}\frac{\partial^2 \mathbf{H}}{\partial t^2} \qquad (36b)}$$

These are wave equations. Their solution has the form of a plane wave

$$\mathbf{E}(\mathbf{r};t) = E_0 \cos(\omega t \pm \mathbf{k}\cdot\mathbf{r}) \qquad (37)$$

Where

$$\mathbf{k} = \frac{\omega}{v}\mathbf{u} \qquad (38)$$

is the **wave vector**. $\mathbf{u}$ is a unit vector that indicates the direction of propagation.

Eqns. (36) can also be written for the individual Cartesian components x, y, z

Time-harmonic fields

Use phasors for time-harmonic (steady-state sinusoidal) fields, e.g.:

$$\mathbf{E}(\mathbf{r};t) = Re\left[\mathbf{E}(\mathbf{r})e^{j\omega t}\right]$$

Maxwell's equations become, for source-free regions:

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} \qquad (39a)$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} \qquad (39b)$$

Which yield wave equations for the phasors:

$$\boxed{\begin{aligned} \nabla^2\mathbf{E} + k^2\mathbf{E} = 0 \qquad (40\text{a}) \\ \nabla^2\mathbf{H} + k^2\mathbf{H} = 0 \qquad (40\text{b}) \end{aligned}}$$

These are called **Helmholtz equations**.

Guided propagation

Let us now assume that there is a structure (e.g. a tube) that guides electromagnetic waves in one direction, z. We write:

$$\mathbf{E}(x, y, z; t) = Re\left[\mathbf{E}(x, y) e^{j(\omega t - \beta z)}\right] \qquad (41)$$

i.e. we describe a wave where the electric field has some dependence on x and y (to be calculated), and depends on z according to a phase factor β (also to be determined)

Note: β is not necessarily the same as k. The presence of the guiding structure can allow for the two to be different. The way to think of them is:

- k is only a property of the medium. It's just $2\pi/\lambda$ for a wave that propagates freely, at speed $v = \frac{1}{\sqrt{\mu\epsilon}}$ in a space filled by the medium having μ and ϵ.
- β is a property of the medium + the guiding structure. We need to solve the Helmholtz and Maxwell equations for the specific structure to find out the relation between β and k.

Transverse and longitudinal components

The phasors $\mathbf{E}(x,y,z)$ and $\mathbf{H}(x,y,z)$ can be divided into transverse and longitudinal components:

$$\mathbf{E}(x,y,z) = [\mathbf{E}_\perp(x,y) + \mathbf{u}_z E_z(x,y)\,]e^{-j\beta z} \qquad (42a)$$

$$\mathbf{H}(x,y,z) = [\mathbf{H}_\perp(x,y) + \mathbf{u}_z H_z(x,y)\,]e^{-j\beta z} \qquad (42b)$$

Substituting (42a,b) into the Maxwell equations (39a,b) we obtain:

$$\frac{\partial E_z}{\partial y} + j\beta E_y = -j\omega\mu H_x \qquad (43a)$$

$$-j\beta E_x - \frac{\partial E_z}{\partial x} = -j\omega\mu H_y \qquad (43b)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z \qquad (43c)$$

$$\frac{\partial H_z}{\partial y} + j\beta H_y = j\omega\epsilon E_x \qquad (43d)$$

$$-j\beta H_x - \frac{\partial H_z}{\partial x} = j\omega\epsilon E_y \qquad (43e)$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega\epsilon E_z \qquad (43f)$$

Transverse components

We can solve for the transverse components as a function of the longitudinal ones. For example, H_x can be written as a function of E_z and H_z by replacing, in (43a), E_y obtained from (43e):

$$H_x = \frac{j}{k_c^2}\left(\omega\epsilon\frac{\partial E_z}{\partial y} - \beta\frac{\partial H_z}{\partial x}\right) \qquad (44a)$$

$$H_y = \frac{-j}{k_c^2}\left(\omega\epsilon\frac{\partial E_z}{\partial x} + \beta\frac{\partial H_z}{\partial y}\right) \qquad (44b)$$

$$E_x = \frac{-j}{k_c^2}\left(\beta\frac{\partial E_z}{\partial x} + \omega\mu\frac{\partial H_z}{\partial y}\right) \qquad (44c)$$

$$E_y = \frac{j}{k_c^2}\left(-\beta\frac{\partial E_z}{\partial y} + \omega\mu\frac{\partial H_z}{\partial x}\right) \qquad (44d)$$

Where

$$k_c = \sqrt{k^2 - \beta^2} \qquad (45)$$

is the **cutoff wavenumber**

Transverse Electric & Magnetic (TEM) waves

Is there a solution of Eqns. (44) where $E_z = H_z = 0$?

Yes, if $k_c = 0$, i.e. $\beta = k$.

From Eqns. (44) we would get an undetermined result, but we can go back to Eqns. (43) and, imposing $E_z = H_z = 0$, find for example:

$$\beta^2 E_y = \omega^2 \mu \epsilon E_y$$

Which means $\beta = \omega\sqrt{\mu\epsilon} = k$ as anticipated

The Helmholtz equation (40a) becomes (take the example for E_x):

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right) E_x = 0 \qquad (46)$$

However we know from (42a) that $E_x(x, y, z) = E_x(x, y)e^{-j\beta z}$, so $\frac{\partial^2}{\partial z^2} E_x = -\beta^2 E_x = -k^2 E_x$ and Eq. (46) becomes:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) E_x = 0 \qquad (47)$$

TEM waves

The same holds of all other transverse components, so we can write

$$\boldsymbol{\nabla}_{\perp}{}^{2}\mathbf{E}_{\perp}(x,y) = 0$$

$$\boldsymbol{\nabla}_{\perp}{}^{2}\mathbf{H}_{\perp}(x,y) = 0$$

Where $\boldsymbol{\nabla}_{\perp}{}^{2} = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

Thus the transverse components of the fields obey Laplace equation, i.e. they are the same as the fields between the conductors.

TEM waves cannot exist in single-conductor waveguides.

This is because, in the perpendicular plane,
$\oint \mathbf{H}_{\perp}(x,y) \cdot d\mathbf{l} = \iint \boldsymbol{\nabla} \times \mathbf{H}_{\perp}(x,y)\, d\mathbf{S} = \iint (J_z + \partial D_z/dt)\, d\mathbf{S}$

But, without an inner conductor, $J_z = 0$.

Moreover, since the TEM mode has by definition $E_z = 0$, then also $D_z = 0$.

$\Rightarrow$ You need a centre conductor to produce a current density $J_z \neq 0$ so that a nonzero component of $\mathbf{H}_{\perp}$ can be produced around it (Ampere law)

Transverse Electric (TE) waves

TE stands for "transverse electric". These are the solutions of Eqns. (44) where $E_z = 0$ whereas $H_z \neq 0$. Eqns. (44) thus become:

$$H_x = -\frac{j\beta}{k_c^2}\frac{\partial H_z}{\partial x} \qquad (48a)$$

$$H_y = -\frac{j\beta}{k_c^2}\frac{\partial H_z}{\partial y} \qquad (48b)$$

$$E_x = -\frac{j\omega\mu}{k_c^2}\frac{\partial H_z}{\partial y} \qquad (48c)$$

$$E_y = \frac{j\omega\mu}{k_c^2}\frac{\partial H_z}{\partial x} \qquad (48d)$$

Now we have solutions where $k_c \neq 0$.

To apply Eqns. (48) we first solve the Helmholtz equation for H_z:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right) H_z = 0 \quad \rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right) H_z = 0$$

because $\frac{\partial^2}{\partial z^2} H_z = -\beta^2 H_z$ and $k_c^2 = k^2 - \beta^2$

Transverse Magnetic (TM) waves

TM stands for "transverse magnetic". These are the solutions of Eqns. (44) where $H_z = 0$ whereas $E_z \neq 0$. Eqns. (44) thus become:

$$H_x = \frac{j\omega\epsilon}{k_c^2}\frac{\partial E_z}{\partial y} \qquad (49a)$$

$$H_y = -\frac{j\omega\epsilon}{k_c^2}\frac{\partial E_z}{\partial x} \qquad (49b)$$

$$E_x = -\frac{j\beta}{k_c^2}\frac{\partial E_z}{\partial x} \qquad (49c)$$

$$E_y = -\frac{j\beta}{k_c^2}\frac{\partial E_z}{\partial y} \qquad (49d)$$

The Helmholtz equation for E_z becomes:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)E_z = 0 \quad \rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right)E_z = 0$$

Wave impedance

In the "circuit style" description of transmission lines, we defined the characteristic impedance as $Z_0 = \frac{V_0^+}{I_0^+} = \sqrt{\frac{L}{C}}$

With single-conductor waveguides this becomes meaningless. We therefore use the wave impedance, defined as the ratio between the transverse components of the electric and magnetic field. It applies to any transmission line.

- **TEM modes**:

using (43d):

$$\boxed{Z_{\mathrm{TEM}} = \frac{E_x}{H_y} = \frac{\omega\mu}{\beta} = \sqrt{\frac{\mu}{\epsilon}} \equiv Z_0 \qquad (50)}$$

where we have used the fact that, for TEM modes, $\beta = k = \omega\sqrt{\mu\epsilon}$

This is the same as Z_0 for a coax (see Tutorial 1).

If μ and ϵ are frequency-independent (non-dispersive medium), then also $\boldsymbol{Z_{\mathrm{TEM}}}$ **is frequency-independent**.

Wave impedance

For non-TEM modes, $\beta = \sqrt{k^2 - k_c^2} \neq k$.

We define the **cutoff (angular) frequency** as:

$$\boxed{\omega_c = \frac{k_c}{\sqrt{\mu\epsilon}} = k_c v \qquad (51)}$$

- **TE modes**:

taking the ratio of (48c) and (48b), and recalling that $k = \omega\sqrt{\mu\epsilon}$:

$$Z_{\text{TE}} = \frac{E_x}{H_y} = \frac{\omega\mu}{\beta} = \frac{\omega\mu}{\sqrt{k^2 - k_c^2}} = \frac{\omega\mu}{\sqrt{\mu\epsilon(\omega^2 - {\omega_c}^2)}} = \omega\sqrt{\frac{\mu}{\epsilon}}\frac{1}{\sqrt{\omega^2 - {\omega_c}^2}}$$

$$\boxed{Z_{\text{TE}} = \frac{Z_0}{\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}} \qquad (52)}$$

$\boldsymbol{Z_{\text{TE}}}$ **is frequency-dependent** and always higher than Z_0

Wave impedance

- **TM modes**:

taking the ratio of (49c) and (49b), and recalling that $k = \omega\sqrt{\mu\epsilon}$:

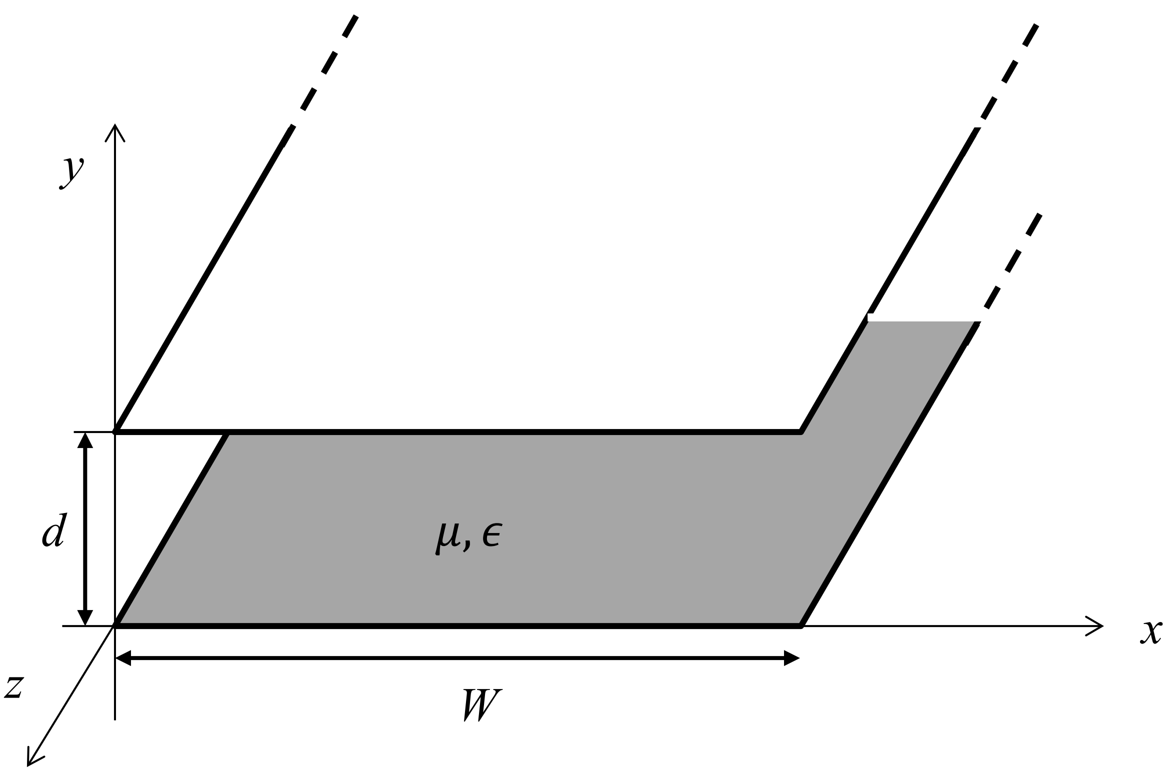
$$Z_{\mathrm{TM}} = \frac{E_x}{H_y} = \frac{\beta}{\omega\epsilon} = \frac{\sqrt{k^2 - k_c^2}}{\omega\epsilon} = \frac{\sqrt{\mu\epsilon(\omega^2 - {\omega_c}^2)}}{\omega\epsilon} = \frac{1}{\omega}\sqrt{\frac{\mu}{\epsilon}}\sqrt{\omega^2 - {\omega_c}^2}$$

$$\boxed{Z_{\mathrm{TM}} = Z_0\sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} \qquad (53)}$$

$\boldsymbol{Z_{\mathrm{TM}}}$ **is frequency-dependent** and always lower than Z_0

Parallel plate waveguide

Consider two metal plates facing each other, separated by a distance d, filled by a medium described by μ and ϵ. The plates have width $W \gg d$ such that fringing fields can be ignored. This parallel plate waveguide is an abstract idealization, but a simple one to analyse.



TM modes

TM modes have $H_z = 0$ and $E_z \neq 0$. Here, in addition, since $W \gg d$ we can consider the waveguide to be invariant along the x direction, thus $\frac{\partial E_z}{\partial x} = 0$.

The Helmholtz equation is:

$$\left(\frac{\partial^2}{\partial y^2} + k_c^2\right) E_z = 0 \qquad (54)$$

and $E_z(x, y, z) = E_z(x, y) e^{-j\beta z}$.

The solution to Eq. (54) is of the form:

$$E_z(x, y) = A \sin(k_c y) + B \cos(k_c y) \qquad (55)$$

where the constants A and B are determined by imposing the boundary conditions $E_z(x, y = 0) = 0$ and $E_z(x, y = d) = 0$ (no electric field along a metal plate).

Therefore $B = 0$ and $k_c d = n\pi$ for $n = 0,1,2,3, \ldots$ or:

$$\boxed{k_{cn} = \frac{n\pi}{d} \qquad (56)} \qquad n = 0,1,2,3, \ldots$$

TM modes

From Eq. (56):

$$\beta = \sqrt{k^2 - k_c^2} \quad \rightarrow \quad \boxed{\beta_n = \sqrt{k^2 - (n\pi/d)^2} \qquad (57)}$$

$$E_z(x, y) = A_n \sin\left(\frac{n\pi y}{d}\right)$$

$$E_z(x, y, z) = A_n \sin\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z} \qquad (58)$$

From Eqns. (49) we find the transverse components:

$$H_x = \frac{j\omega\epsilon}{k_c} A_n \cos\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z} \qquad (59a)$$

$$E_y = -\frac{j\beta_n}{k_c} A_n \cos\left(\frac{n\pi y}{d}\right) e^{-j\beta_n z} \qquad (59b)$$

$$E_x = H_y = 0 \qquad (59c)$$

From Eq. (51), the cutoff frequency for the *n*-th mode is:

$$\boxed{f_{cn} = \frac{k_{cn}}{2\pi\sqrt{\mu\epsilon}} = \frac{n}{2d\sqrt{\mu\epsilon}} \qquad (60)}$$

TM modes

We can also define a cutoff wavelength for mode n:

$$\lambda_{cn} = \frac{v}{f_{cn}} = \frac{1/\sqrt{\mu\epsilon}}{n/2d\sqrt{\mu\epsilon}} = \frac{2d}{n} \qquad (61)$$

Or:

$$\boxed{d = n\frac{\lambda_{cn}}{2} \qquad (62)}$$

This means that, in order to start propagating the n-th mode, you must be able to fit at least n half-wavelengths between the plates.

Bouncing plane waves interpretation

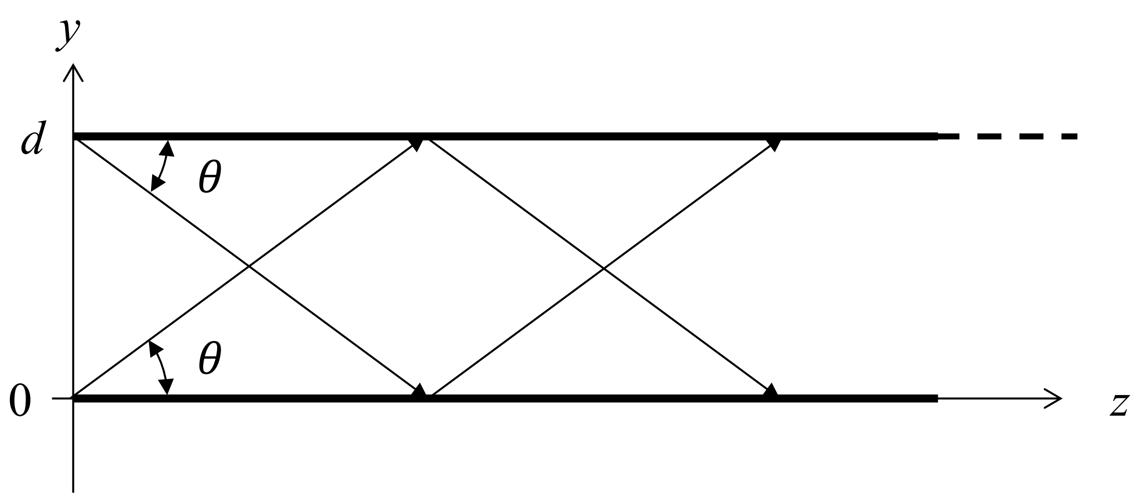
Take for example the TM_1 mode ($n = 1$):

$$\beta_1 = \sqrt{k^2 - (\pi/d)^2} = \sqrt{k^2 - (2\,\pi/\lambda_{c1})^2}\,, \qquad E_z(x,y,z) = A_1 \sin\left(\frac{\pi y}{d}\right) e^{-j\beta_1 z}.$$

This can be rewritten as:

$$E_z(x,y,z) = \frac{A_1}{2j}\left\{e^{j(\pi y/d\, - \beta_1 z)} - e^{j(-\pi y/d - \beta_1 z)}\right\}$$

Interpretation: two plane waves travelling along a oblique directions $(-y, +z)$ and $(+y, +z)$, with wave vector $k = \sqrt{(\pi/d)^2 + {\beta_1}^2}$



$$k\ \sin\theta = \frac{\pi}{d}$$

$$k\ \cos\theta = \beta_1$$

Phase velocity

The phase velocity of each plane wave along its direction of propagation (θ) is simply $\omega/k = 1/\sqrt{\mu\epsilon}$ which is just the speed of light in the material filling the guide.

BUT, the phase velocity along z is:

$$v_z = \frac{\omega}{\beta_1} = \frac{1}{\cos\theta\sqrt{\mu\epsilon}} \qquad (63)$$

which is always greater than the speed of light in the material!!

This already tells you that the phase velocity is not the quantity describing the speed at which information travels along a transmission line.

TE modes

The TE modes in a parallel plate waveguide are dual to the TM modes:

$$H_z(x,y) = A\sin(k_c y) + B\cos(k_c y) \qquad (64)$$

From (48c):

$$E_x(x,y,z) = -\frac{j\omega\mu}{k_c}[A\cos(k_c y) - B\sin(k_c y)]e^{-j\beta z}$$

with $E_x = 0$ at $y = 0, d$ and $E_z \equiv 0 \;\Rightarrow A = 0$

$$H_z(x,y,z) = B_n\cos\left(\frac{n\pi y}{d}\right)e^{-j\beta_n z} \qquad (65)$$

$$E_x = \frac{j\omega\mu}{k_c}B_n\sin\left(\frac{n\pi y}{d}\right)e^{-j\beta_n z} \qquad (66a)$$

$$H_y = \frac{j\beta_n}{k_c}B_n\sin\left(\frac{n\pi y}{d}\right)e^{-j\beta_n z} \qquad (66b)$$

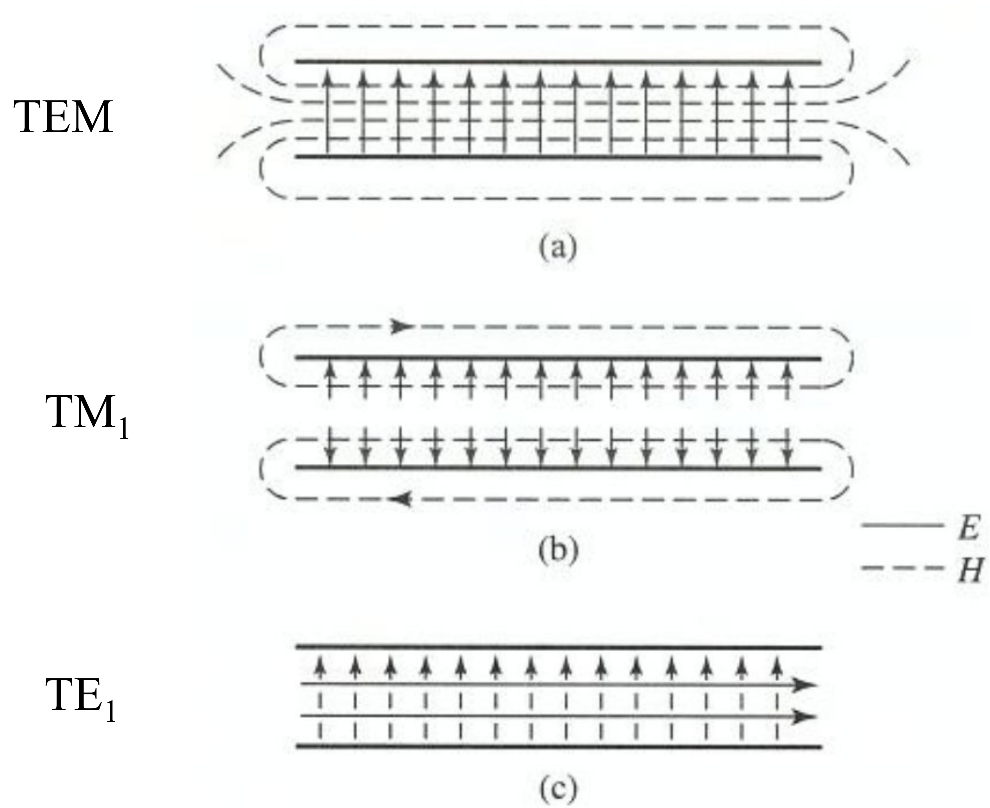
$$E_y = H_x = 0 \qquad (66c)$$

β_n and f_{cn} are the same as for the TM mode.

TEM mode

Since the parallel plate waveguide has two separate conductors, it can actually sustain a TEM mode, which is just the TM mode with $n = 0$.

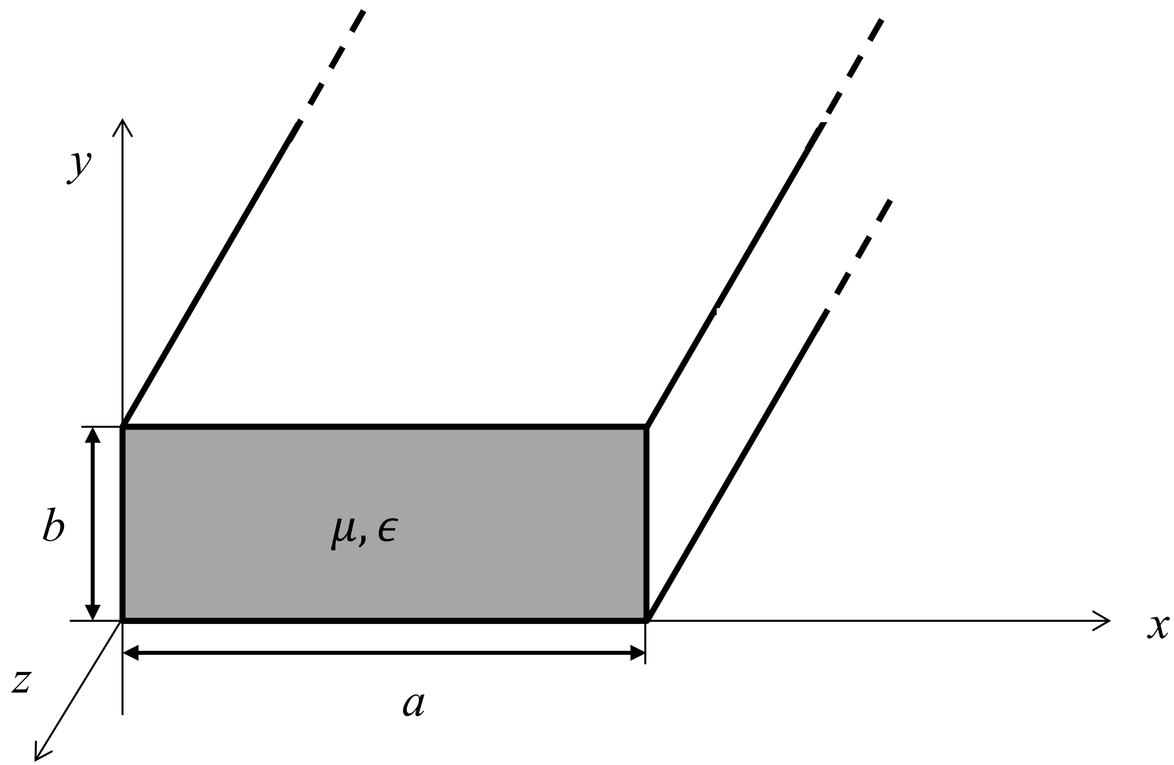
$$\beta_0 = k; \quad f_{c0} = 0; \quad H_x = \frac{j\omega\epsilon}{k_c} A_0 \, e^{-j\beta_0 z}; \quad E_y = -\frac{j\beta_n}{k_c} A_0 \, e^{-j\beta_0 z}$$



Rectangular waveguides

Rectangular waveguides are among the most commonly used in high-frequency microwave applications. Since there is now only one conductor, the TEM mode does not exist.

By convention
$a > b$



TM modes

Helmholtz equation: (now $\partial/\partial x \neq 0$ because it's not invariant along x)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + k_c^2\right) E_z(x,y) = 0; \qquad E_z(x,y,z) = E_z(x,y)e^{-j\beta z}$$

This partial differential equation can be solved by separating the variables:

$$E_z(x,y) = X(x)Y(y)$$

Substituting into the Helmholtz equation:

$$Y\frac{\partial^2 X}{\partial x^2} + X\frac{\partial^2 Y}{\partial y^2} + k_c^2 XY = 0 \qquad \rightarrow \qquad \frac{1}{X}\frac{\partial^2 X}{\partial x^2} + \frac{1}{Y}\frac{\partial^2 Y}{\partial y^2} + k_c^2 = 0$$

Since the variables are independent, each term must be equal to a constant. We can define the constants k_x and k_y such that:

$$\frac{\partial^2 X}{\partial x^2} + k_x^2 X = 0 \qquad (67a)$$

$$\frac{\partial^2 Y}{\partial y^2} + k_y^2 Y = 0 \qquad (67b)$$

$$k_x^2 + k_y^2 = k_c^2 \qquad (67c)$$

TM modes

The general solution for E_z can be written as:

$$E_z(x,y) = [A\cos(k_x x) + B\sin(k_x x)]\big[C\cos(k_y y) + D\sin(k_y y)\big]$$

Boundary conditions:

$E_z(x,y) = 0 \quad$ at $\quad x = 0, a$

$E_z(x,y) = 0 \quad$ at $\quad y = 0, b$

They impose:

$$A = 0; \qquad k_{xm} = \frac{m\pi}{a} \qquad m = 1,2,3,\ldots$$

$$C = 0; \qquad k_{yn} = \frac{n\pi}{b} \qquad n = 1,2,3,\ldots$$

$$E_z(x,y,z) = B_{mn}\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta z} \qquad (68)$$

The propagation constant is:

$$\boxed{\beta_{mn} = \sqrt{k^2 - {k_{cmn}}^2} = \sqrt{k^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2} \qquad (69)}$$

TM modes – cutoff frequency

From (69) we see that the propagation constant is real if $k > k_{cmn}$, where the cutoff frequency of the TM_{mn} mode is:

$$\boxed{f_{cmn} = \frac{k_{cmn}}{2\pi\sqrt{\mu\epsilon}} = \frac{1}{2\pi\sqrt{\mu\epsilon}}\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \qquad (70)}$$

TM modes: transverse components

From (69) and (49), the transverse components of the fields in the TM_{mn} mode are:

$$E_x = -\frac{j\beta_{mn} m\pi}{a{k_c}^2} B_{mn} \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn} z} \quad (71a)$$

$$E_y = -\frac{j\beta_{mn} n\pi}{b{k_c}^2} B_{mn} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn} z} \quad (71b)$$

$$H_x = \frac{j\omega\epsilon n\pi}{b{k_c}^2} B_{mn} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn} z} \quad (71c)$$

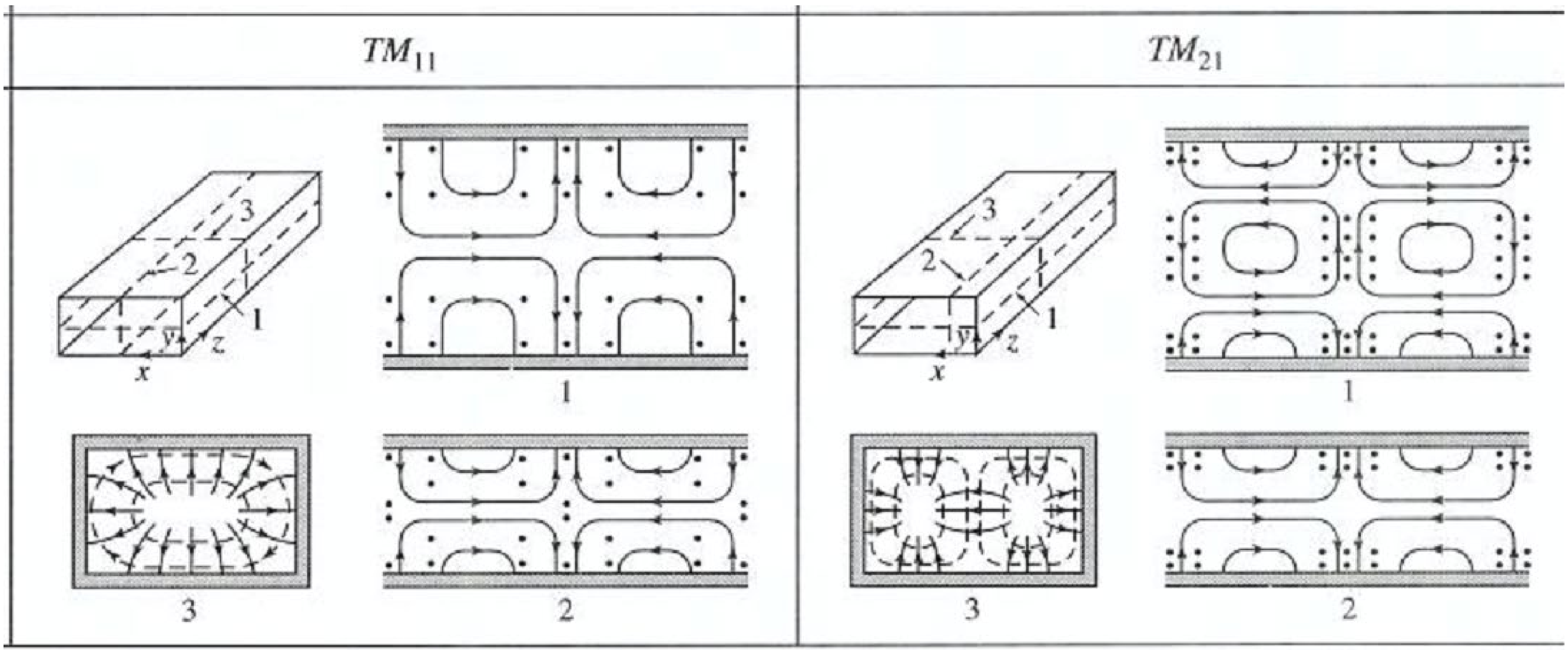
$$H_y = -\frac{j\omega\epsilon m\pi}{a{k_c}^2} B_{mn} \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn} z} \quad (71d)$$

Notice that all fields are identically zero unless **both** m and n are $\neq 0$.

Therefore, **the lowest propagating TM mode is the $\mathbf{TM_{11}}$**, with cutoff

$$f_{c11} = \frac{1}{2\sqrt{\mu\epsilon}} \sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2}$$

TM modes: field patterns



TE modes

The solution is obtained similarly to the TM modes:

$$H_z(x,y,z) = A_{mn}\cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z} \quad (72)$$

$$E_x = \frac{j\omega\mu n\pi}{bk_c{}^2}A_{mn}\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta_{mn}z} \quad (73a)$$

$$E_y = -\frac{j\omega\mu m\pi}{ak_c{}^2}A_{mn}\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta_{mn}z} \quad (73b)$$

$$H_x = \frac{j\beta m\pi}{ak_c{}^2}A_{mn}\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta_{mn}z} \quad (73c)$$

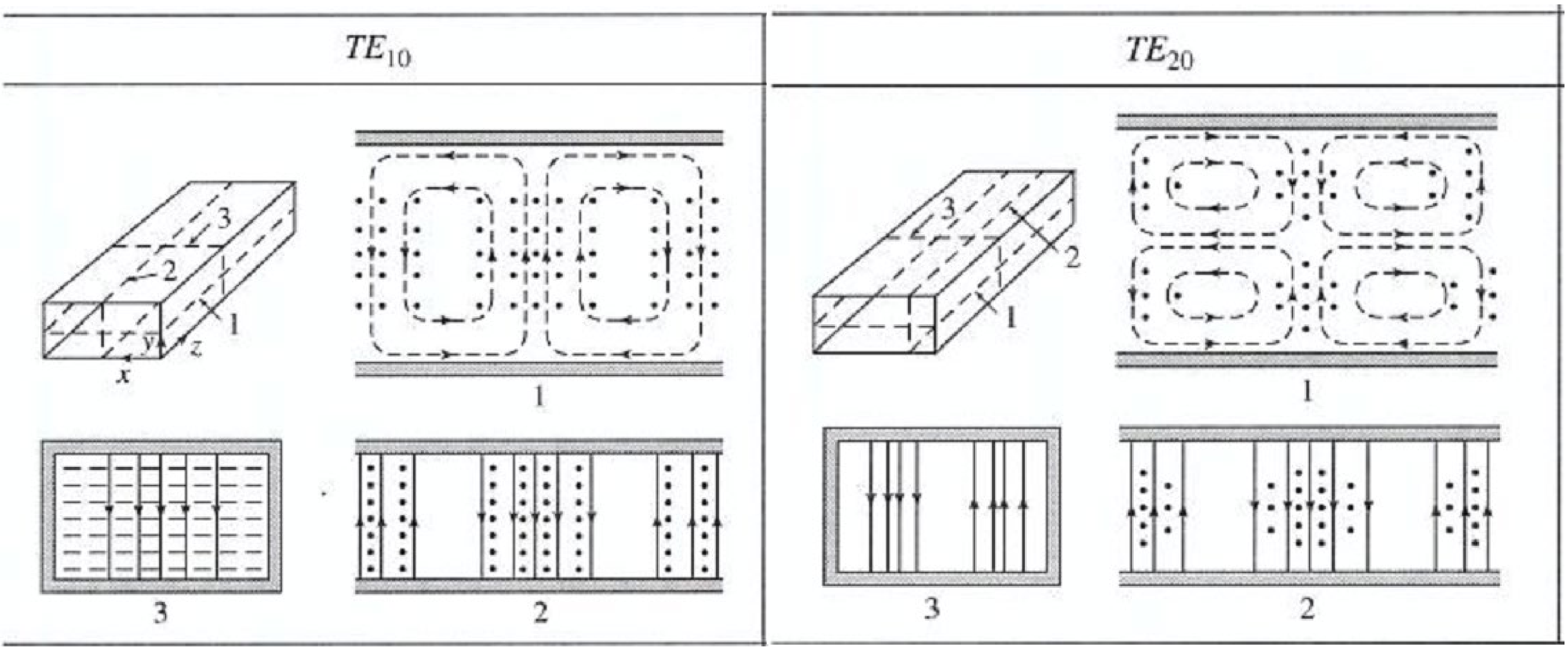
$$H_y = \frac{j\beta n\pi}{bk_c{}^2}A_{mn}\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta_{mn}z} \quad (73d)$$

Now the fields are nonzero as soon as **either** m or n is $\neq 0$.

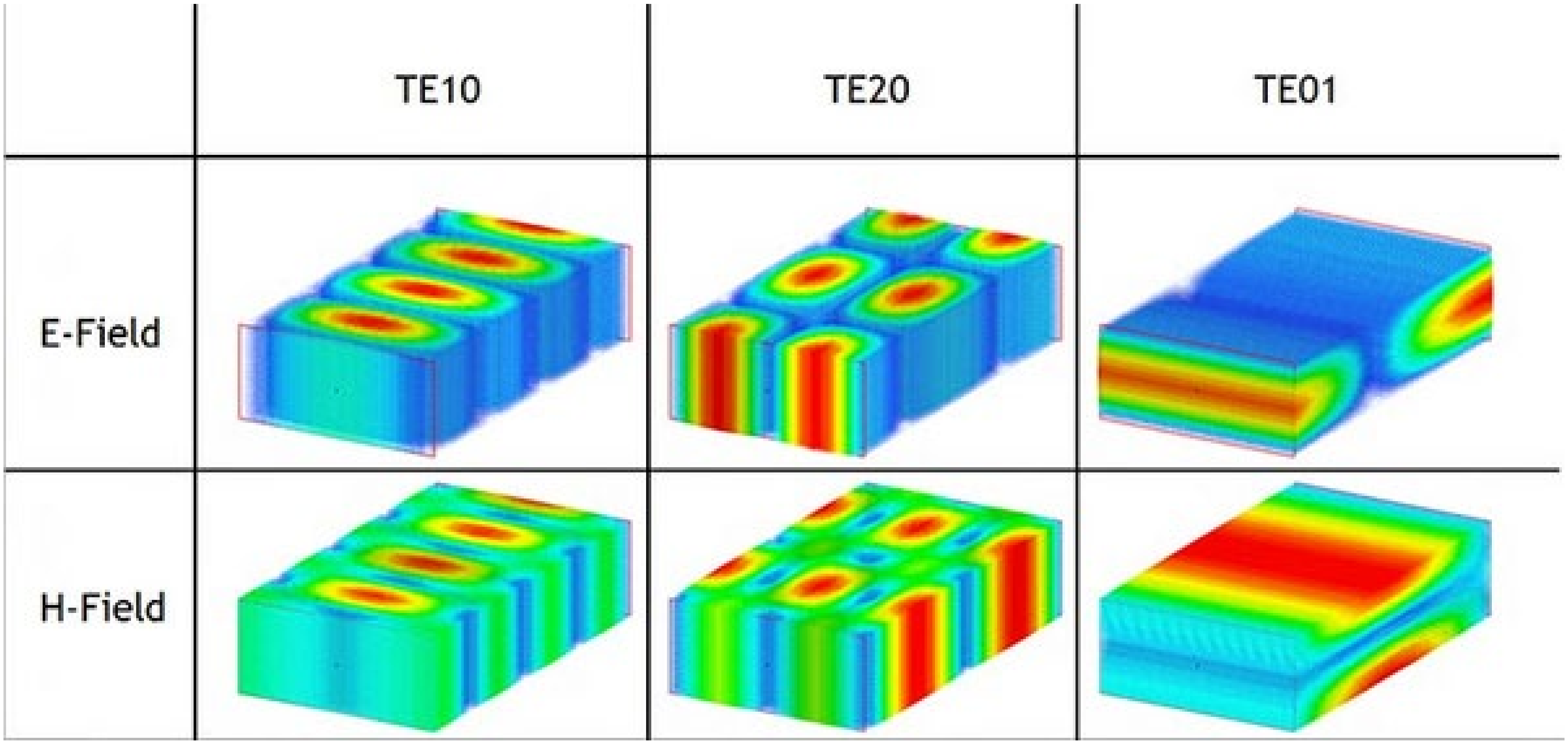
Therefore, **the lowest propagating TE mode is the TE$_{10}$**, which has:

$$\boxed{f_{c10} = \frac{1}{2a\sqrt{\mu\epsilon}} \quad (74)}$$

TE modes: field patterns



TE modes: field patterns



Waveguide modes: choice

The "dominant" (i.e. starting at lowest frequency) mode is always the TE_{10}.

Other modes become accessible at higher frequencies.

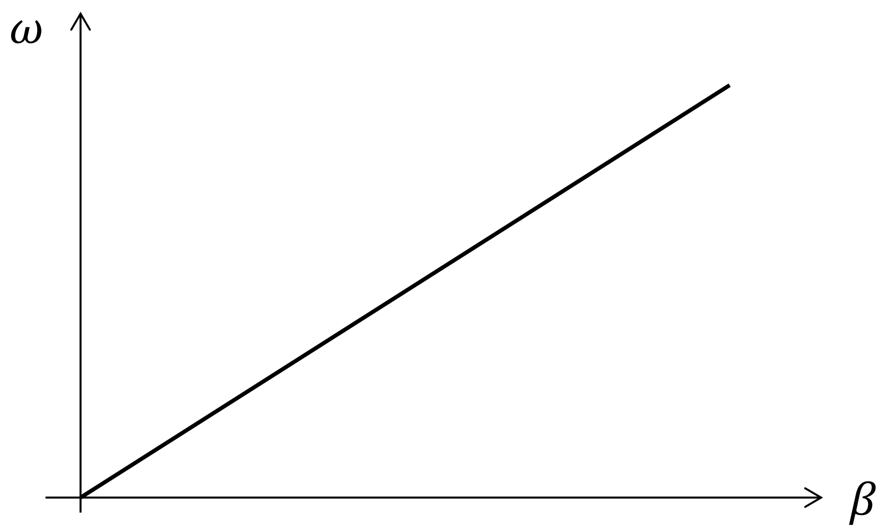
If you want to transmit a frequency f_s, you will normally choose a waveguide such that $f_{c10} < f_s < f_{c01} < f_{c11}$ ... because that's the smallest waveguide you can use for the purpose. Waveguides are bulky and heavy, so you don't normally want to make them bigger than necessary just to propagate higher modes!

Dispersion

In empty space or in a uniform medium, electromagnetic waves propagate with:

- velocity $v = 1/\sqrt{\mu\epsilon}$
- propagation constant $\beta = \omega/v$
- wavelength $\lambda = 2\pi/\beta = v/f$

If μ and ϵ are independent of frequency, then v is constant, and β and ω are simply proportional to each other

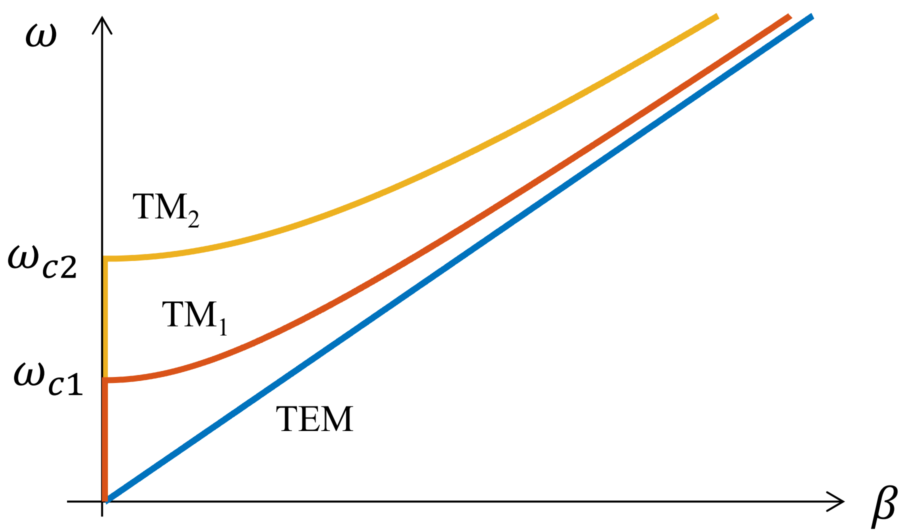


The function $\omega(\beta)$ is called **dispersion relation**
If it's linear, the medium is called non-dispersive, because it means that the velocity of propagation is the same for signals of any frequency.

Dispersion

Consider the TM modes in a parallel-plate waveguide:

- $\mathrm{TM}_0 \equiv \mathrm{TEM}$: $\beta_0 = k = \omega/v, \quad v = 1/\sqrt{\mu\epsilon}, \quad \omega_{c0} = 0,$
- TM_1: $\beta_1 = \sqrt{k^2 - {k_{c1}}^2} = \frac{\omega}{v}\sqrt{1 - \left(\frac{\omega_{c1}}{\omega}\right)^2} \quad (\omega_{c1} = k_{c1}v, k_{c1} = \pi/d)$
- TM_2: $\beta_2 = \sqrt{k^2 - {k_{c2}}^2} = \frac{\omega}{v}\sqrt{1 - \left(\frac{\omega_{c2}}{\omega}\right)^2} \quad (\omega_{c2} = k_{c2}v, k_{c2} = 2\pi/d)$

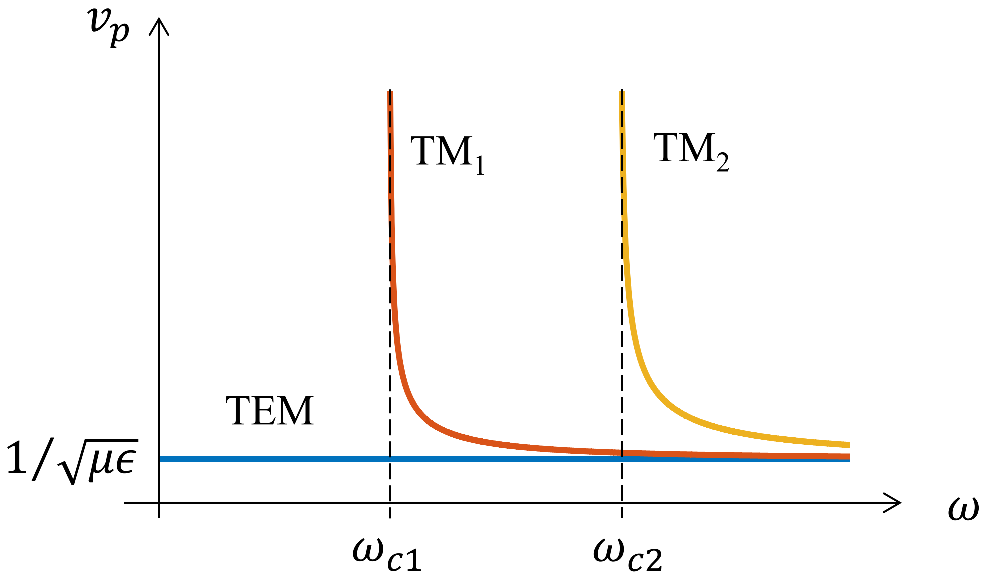


For $n \geq 1$ the modes are **dispersive**:
the relation $\omega(\beta)$ is not linear

Phase velocity

The phase velocity is, in general, given by:

$$v_p(\omega) = \frac{\omega}{\beta(\omega)} \quad (75)$$



For $n \geq 1$ the phase velocity is infinite at the cutoff frequency, and tends asymptotically to the speed of light in the medium as $\omega \to \infty$

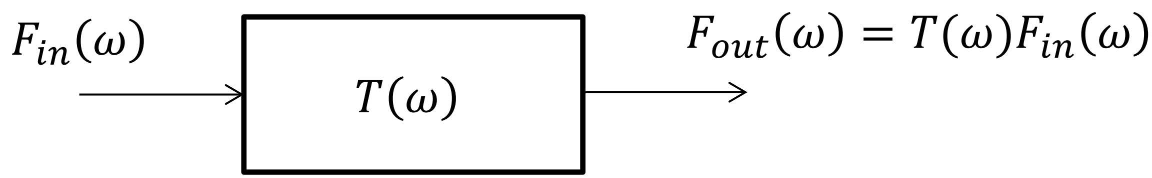
Time delay

A transmission line can be treated like a two-port network, with transfer function $T(\omega)$. We apply a signal $f_{in}(t)$ to the input of the line. Its Fourier transform is:

$F_{in}(\omega) = \int_{-\infty}^{+\infty} f_{in}(t)\, e^{-j\omega t} dt$ and the inverse is $f_{in}(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F_{in}(\omega)\, e^{j\omega t} d\omega$

If the TL is matched and lossless, then

$$T(\omega) = e^{-j\beta z} = e^{-j\frac{\omega}{v_p} z} \qquad (76)$$

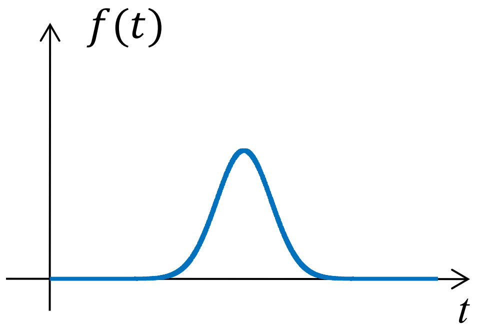


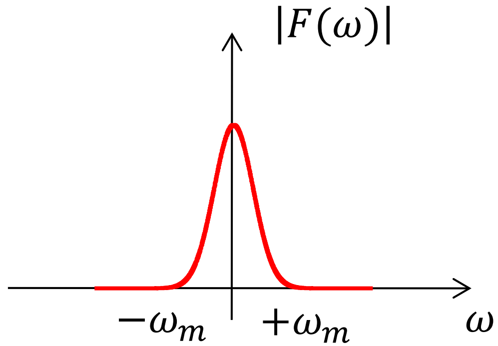
$$f_{out}(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F_{in}(\omega)\, e^{j(\omega t - \beta z)} d\omega = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F_{in}(\omega)\, e^{j\omega\left(t - \frac{z}{v_p}\right)} d\omega = f_{in}\left(t - \frac{z}{v_p}\right)$$

If v_p is constant (independent of ω), the line puts out an exact copy of the input signal, delayed by a time z/v_p

Narrow-band signal

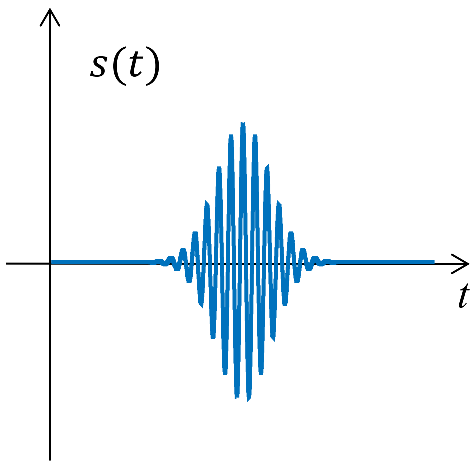
Consider a signal $f(t)$ having a narrow band, i.e. its Fourier transform has significant intensity only between $\pm\omega_m$.

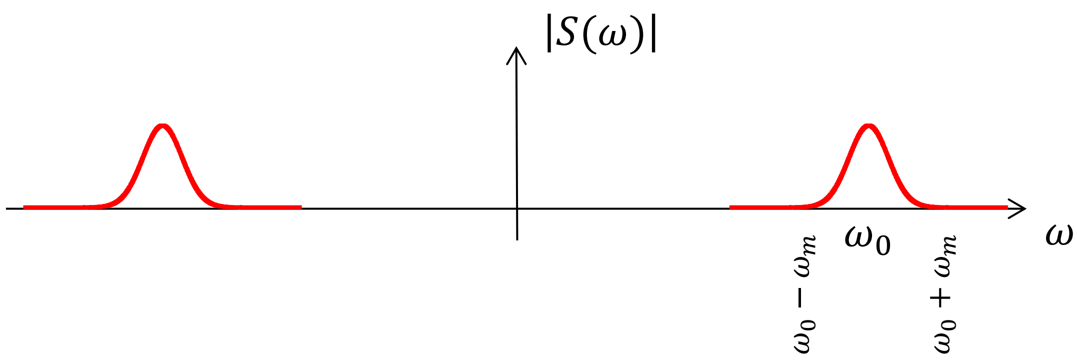




The signal $f(t)$ then modulates a carrier of the form $\cos(\omega_0 t)$

$$s(t) = f(t)\cos(\omega_0 t) = Re\{f(t)e^{j\omega_0 t}\}$$





Narrow-band signal

The Fourier transform of the modulated signal is:

$$S(\omega) = \int_{-\infty}^{+\infty} f(t)\, e^{j\omega_0 t} e^{-j\omega t} dt = F(\omega - \omega_0)$$

(I have taken only the positive frequencies part. When I go back to the time domain, I will take the real part of the inverse Fourier transform of the above)

Now pass this signal through a TL with transfer function:

$$T(\omega) = e^{-j\beta(\omega)z}$$

i.e. I am now assuming a dispersive TL.

$$S_{out}(\omega) = F(\omega - \omega_0) e^{-j\beta(\omega)z}$$

Back to the time domain:

$$s_{out}(t) = \frac{1}{2\pi} Re \int_{-\infty}^{+\infty} S_{out}(\omega)\, e^{j\omega t} d\omega =$$

$$= \frac{1}{2\pi} Re \int_{\omega_0 - \omega_m}^{\omega_0 + \omega_m} F(\omega - \omega_0)\, e^{j(\omega t - \beta(\omega) z)} d\omega \qquad (77)$$

Narrow-band signal

Although $\beta(\omega)$ is no longer constant, I can take a Taylor expansion of it around ω_0 because the signal is narrow-band around it

$$\beta(\omega) \approx \beta_0 + \beta'(\omega - \omega_0) + \ \dots$$

where $\beta_0 = \beta(\omega_0)$, $\quad \beta' = \left.\frac{d\beta}{d\omega}\right|_{\omega=\omega_0}$

Substituting into eq. (77) and making a change of variable $y = \omega - \omega_0$:

$$s_{out}(t) = \frac{1}{2\pi} Re \int_{-\omega_m}^{\omega_m} F(y)\, e^{j\left(\omega_0 t + yt - \beta_0 z - \beta' yz\right)} dy =$$

$$= \frac{1}{2\pi} Re \left\{ e^{j(\omega_0 t - \beta_0 z)} \int_{-\omega_m}^{\omega_m} F(y)\, e^{jy\left(t - \beta' z\right)} dy \right\} =$$

$$= \underbrace{\cos(\omega_0 t - \beta_0 z)}_{(1)} \cdot \underbrace{f(t - \beta' z)}_{(2)} \qquad (78)$$

Group velocity

$$\beta_0 = \beta(\omega_0) = \frac{\omega_0}{v_p(\omega_0)}$$

Term (1) becomes

$$\cos\left[\omega_0\left(t - \frac{z}{v_p(\omega_0)}\right)\right]$$

Let us define a **group velocity**

$$\boxed{v_g(\omega_0) = \frac{1}{\beta'} = \left(\frac{d\beta}{d\omega}\right)^{-1}\Bigg|_{\omega=\omega_0} \qquad (79)}$$

Term (2) becomes

$$f\left(t - \frac{z}{v_g(\omega_0)}\right)$$

The carrier "propagates" at the phase velocity

The envelope propagates at the group velocity

Group velocity in waveguides

Consider an air-filled waveguide, where $k = \omega/c\,;\quad k_0 = \omega_0/c$:

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{(\omega/c)^2 - k_c^2}$$

$$\left.\frac{d\beta}{d\omega}\right|_{\omega=\omega_0} = \frac{\omega_0/c^2}{\sqrt{(\omega_0/c)^2 - k_c^2}} = \frac{k_0}{c\beta}$$

$$v_g(\omega_0) = \left.\left(\frac{d\beta}{d\omega}\right)^{-1}\right|_{\omega=\omega_0} = c\,\frac{\beta}{k_0} \qquad (80)$$

The phase velocity is

$$v_p(\omega_0) = \frac{\omega_0}{\beta} = c\,\frac{k_0}{\beta} \qquad (81)$$

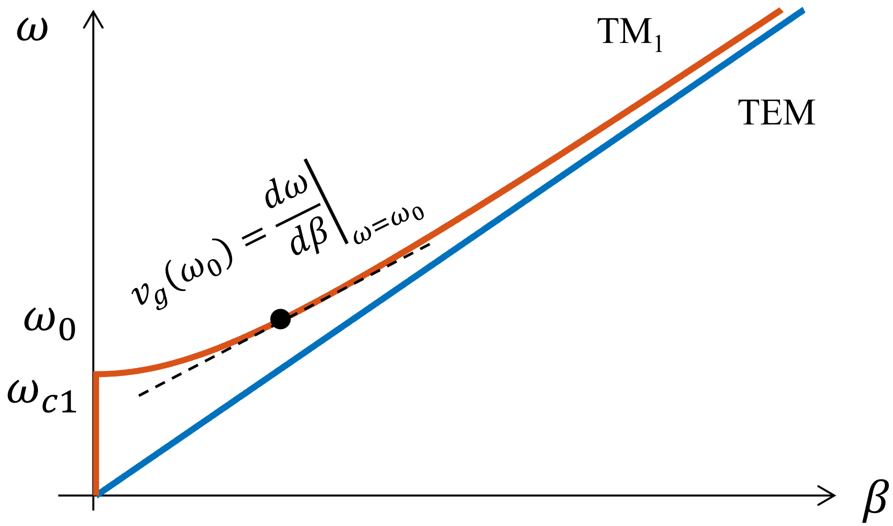
Since $\beta \leq k$ (always!), we have $\beta/k_0 \leq 1$ and $k_0/\beta \geq 1$, and therefore:

$$\boxed{v_g \leq c \leq v_p}$$

The wave packet always propagates at or slower than the speed of light.

Group velocity in waveguides

Graphical interpretation:



The group velocity is the slope of the dispersion relation.

Notice that $v_g \rightarrow 0$ as you approach the cutoff frequency (the dispersion relation tends to horizontal)

Electromagnetic power

Calculate the divergence of the vector product of $\mathbf{E}$ and $\mathbf{H}$.

A useful vector identity allows us to write:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}) \qquad (82)$$

The first term on the right-hand side is, using Faraday's law:

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} \qquad (83)$$

The second term is, using Ampere's law:

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \mathbf{E} \cdot \mathbf{J} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} \qquad (84)$$

In Eq. (83), the second term is only a function of the magnetic fields, and in (84) only a function of the electric fields.

Electromagnetic power

In electrical circuits, the dissipated power is expressed by Joule's law:

$$P = V \cdot I \qquad (85)$$

Therefore, in microscopic terms, the power density is:

$$\frac{dP}{dV} = \mathbf{E} \cdot \mathbf{J} \qquad (86)$$

measured in W/m^3. V is the volume.

Since $\mathbf{E} \cdot \mathbf{J}$ is a power density, so are the terms in Eqns. (83) and (84).

The total power is obtained by integrating (82) over the volume V:

$$\iiint_V \nabla \cdot (\mathbf{E} \times \mathbf{H}) dV = -\iiint_V \left(\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t}\right) dV - \iiint_V \mathbf{E} \cdot \mathbf{J}\, dV \qquad (87)$$

Eq. (84) can be rewritten using Gauss' theorem, and the identities

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\mathbf{E} \cdot \mathbf{D}}{2}\right); \qquad \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\mathbf{H} \cdot \mathbf{B}}{2}\right)$$

where $\mathbf{E} \cdot \mathbf{D} = \epsilon E^2$ and $\mathbf{H} \cdot \mathbf{B} = \mu H^2$

Poynting theorem

$$\boxed{\oiint_S (\mathbf{E} \times \mathbf{H}) dS = -\frac{\partial}{\partial t} \iiint_V \left(\frac{\mu H^2}{2} + \frac{\epsilon E^2}{2} \right) dV - \iiint_V \mathbf{E} \cdot \mathbf{J} \, dV \qquad (88)}$$

where S is the surface of volume V. Eq. (88) is in units of W.

Eq. (88) is called **Poynting theorem**.

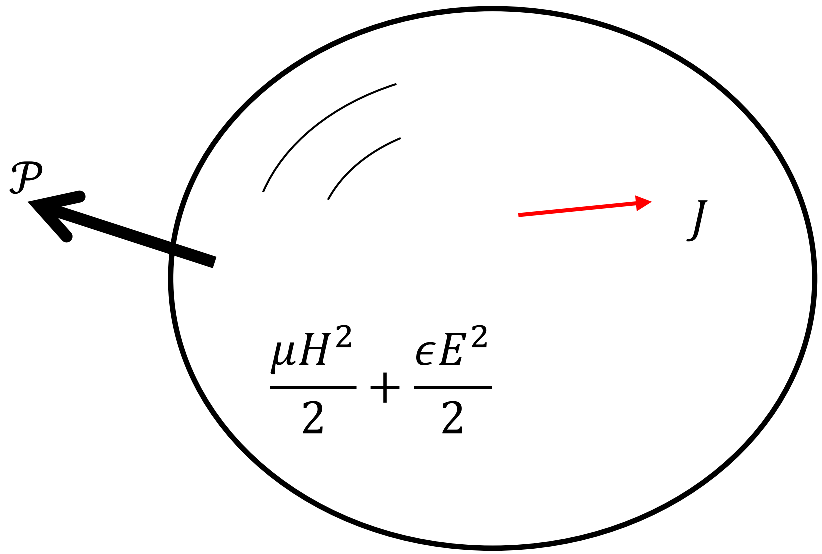
Let us define the **Poynting vector:** (in units W/m^2)

$$\boxed{\mathcal{P} = \mathbf{E} \times \mathbf{H} \qquad (89)}$$

From Eq. (88) we can identify the various terms:

- $\oiint_S (\mathbf{E} \times \mathbf{H}) dS$ is the flux of the Poynting vector out of the volume V
- $\frac{\mu H^2}{2}$ is the density of magnetic power
- $\frac{\epsilon E^2}{2}$ is the density of electric power
- $\mathbf{E} \cdot \mathbf{J}$ is the density of power dissipated by Joule heating

Poynting theorem



The Poynting theorem says that:

"the outward flux of the Poynting vector equals the negative of the time change of electric and magnetic power densities, and the Joule dissipation."

The Poynting vector indicates the direction of propagation of the power associated with an electromagnetic wave

Time-averaged and complex Poynting vector

Typically we deal with time-varying harmonic signals, so it's more meaningful to talk about the Poynting vector averaged over one period of the wave:

$$\mathcal{P}_{av} = \frac{1}{T}\int_0^T \mathcal{P}(t)dt$$

Using the phasor description of $\mathbf{E}$ and $\mathbf{H}$ one can demonstrate that

$$\mathcal{P}_{av} = \frac{1}{2} Re\{\mathbf{E} \times \mathbf{H}^*\} \qquad (90)$$

(it's the same reason why the power in a circuit is $\frac{1}{2} Re\{\mathbf{V}\,\mathbf{I}^*\}$)

We can define a complex Poynting vector as

$$\mathcal{P}_{av} = \mathbf{E} \times \mathbf{H}^* \qquad (91)$$

Power in waveguides

From the equations of the fields in waveguides (59), (66), (71), (73) we can calculate the Poynting vector for any mode, and integrate it over the cross-section of the waveguide to obtain the total power flowing through.

Example: TE modes in rectangular waveguide

$$E_x(x,y,z) = \frac{j\omega\mu n\pi}{bk_{cmn}^2} A_{mn} \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn}z} \quad (73a)$$

$$E_y(x,y,z) = -\frac{j\omega\mu m\pi}{ak_{cmn}^2} A_{mn} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn}z} \quad (73b)$$

$$H_x(x,y,z) = \frac{j\beta m\pi}{ak_{cmn}^2} A_{mn} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn}z} \quad (73c)$$

$$H_y(x,y,z) = \frac{j\beta n\pi}{bk_{cmn}^2} A_{mn} \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) e^{-j\beta_{mn}z} \quad (73d)$$

Power in waveguides

The power is the sum of the contributions from $E_x H_y{}^*/2$ and $-E_y H_x{}^*/2$.
The $-$ sign is because $\hat{\mathbf{y}} \times \hat{\mathbf{x}} = -\hat{\mathbf{z}}$

$$\mathcal{P}_{av}(x,y) = \hat{\mathbf{z}}\frac{1}{2}Re\left\{E_x(x,y)H_y{}^*(x,y) - E_y(x,y)H_x{}^*(x,y)\right\}$$

Notice, however, that E_y has itself a $-$ sign [Eq. (73b)] so all terms give a positive contribution.

The power flow is then $\oiint_S \; \mathcal{P}_{av}(x,y) \cdot d\mathbf{S}$, where $d\mathbf{S} = \hat{\mathbf{z}}\, dxdy$

For example, for the TE_{10} mode ($m = 1, n = 0$) one obtains:

$$P_{10} = \frac{\omega\mu a^3 b |A_{10}|^2}{4\pi^2} Re\{\beta\} \quad (92)$$

Note: we used a generic amplitude A_{mn}. Specifically, one finds that

- $A_{mn} \equiv H_{mn}$ for TE modes
- $A_{mn} \equiv E_{mn}$ for TM modes

Evanescent waves

In a lossless waveguide, the propagation constant is $\gamma = j\beta$, where

$$\beta = \pm\sqrt{k^2 - {k_{cmn}}^2}$$

$$k = \frac{\omega}{v} = \omega\sqrt{\mu\epsilon}$$

$$k_{cmn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2} \equiv \omega_{cmn}\sqrt{\mu\epsilon}$$

Therefore

$$\beta = \pm\omega\sqrt{\mu\epsilon}\sqrt{1 - \left(\frac{\omega_{cmn}}{\omega}\right)^2} \qquad (93)$$

If $\omega > \omega_{cmn}$ then β is real, γ is imaginary, and the wave propagates normally, accumulating a phase factor.

Evanescent waves

If $\omega < \omega_{cmn}$ then β is imaginary, γ is real, and the wave decays exponentially as it goes. We can define an attenuation constant below cutoff as:

$$\alpha_{bc} = \omega\sqrt{\mu\epsilon}\sqrt{\left(\frac{\omega_{cmn}}{\omega}\right)^2 - 1} \qquad (94)$$

Therefore, the electric and magnetic fields in a waveguide below cutoff propagate as (take for example the x-component of the electric field):

$$E_x(x, y, z) = E_x(x, y, 0)e^{-\alpha_{bc}z} \qquad (95)$$

An electromagnetic wave that decays exponentially due to being below the frequency cutoff of a guiding structure is called **evanescent wave**.

Note that the attenuation constant α_{bc} does not arise from dissipation! The waveguide was assumed lossless, so there is no heating taking place anywhere. The power is reflected back at the point where the waveguide goes below cutoff, but it's not sudden.

Units for propagation constant

In general, the propagation constant has a real and an imaginary part:

$$\gamma = \alpha + j\beta$$

Their units are:

$$[\beta] = \mathrm{rad/m}$$

$$[\alpha] = \mathrm{Np/m}$$

Np stands for "Neper". The Neper is a dimensionless unit that describes by how many multiples of e (the Neper constant) the wave gets attenuated.

$\alpha = 1$ Np/m means that the field (electric or magnetic) has amplitude reduced by $1/e$ after 1 m.

Since $20\mathrm{Log}_{10}e = 8.69$ we can also write:

$$\boxed{1\ \mathrm{Np/m} = 8.69\ \mathrm{dB/m}}$$

Waveguides as high-pass filters

Consider for example a waveguide having cutoff for the lowest mode (TE_{10}) $f_{c10} = 10$ GHz.

Calculate the attenuation of a signal at 8 GHz.

$$\alpha_{bc} = 2\pi f\sqrt{\mu\epsilon}\sqrt{\left(\frac{f_{c10}}{f}\right)^2 - 1} = \frac{2\pi\times 8\times 10^9}{3\times 10^8}\sqrt{\left(\frac{10}{8}\right)^2 - 1} = 125.66 \text{ Np/m}$$

Or $125.66 \times 8.69 = 1092$ dB/m

This is an extremely effective high-pass filter!!